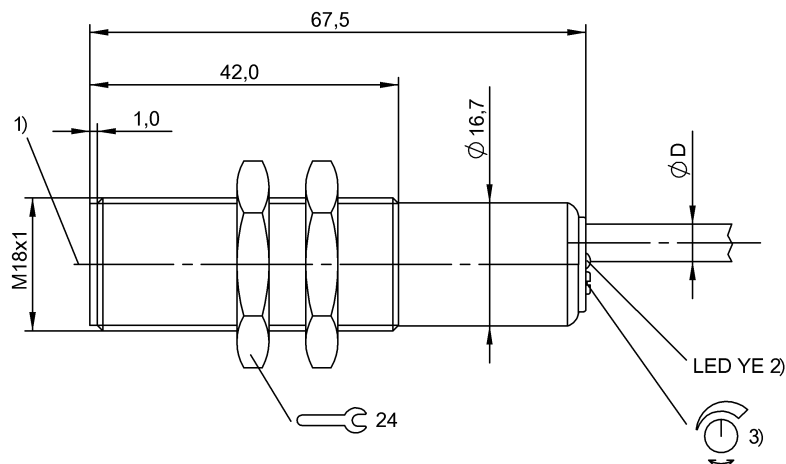




1) Optical axis, 2) Output function, 3) Sn



Basic features

Approval/Conformity	CE cULus
Basic standard	IEC 60947-5-2
Principle of operation	Photoelectric sensor
Series	18M
Style	Cylinder Straight optics

Display/Operation

Adjuster	12-turn potentiometer
Display	Output function- LED yellow
Setting	Rated switching distance (Sn)

Electrical connection

Cable diameter D	4.60 mm
Cable length L	3 m
Conductor cross-section	0.34 mm ²
Connection	Cable, 3.00 m, PVC
Number of conductors	3
Polarity reversal protected	yes
Protection against device mix-ups	yes
Short-circuit protection	yes

Electrical data

Load capacitance max. at U _e	1 µF
No-load current I _o max. at U _e	20 mA
Operating voltage U _b	10...30 VDC
Output resistance R _a	33.0 kOhm
Protection class	II
Rated insulation voltage U _i	250 V AC
Rated operating current I _e	200 mA
Rated operating voltage U _e DC	24 V
Ready delay t _v max.	10 ms
Residual current I _r max.	50 µA
Ripple max. (% of U _e)	10 %
Switching frequency	100 Hz
Turn-off delay t _{off} max.	5 ms
Turn-on delay t _{on} max.	5 ms
Utilization category	DC -13
Voltage drop U _d max. at I _e	2.5 V

Environmental conditions

Ambient temperature	-5...55 °C
Contamination scale	3
EN 60068-2-27, Shock	Half-sinus, 30 g _n , 11 ms, 3x6
EN 60068-2-6, Vibration	10...55 Hz, amplitude 1 mm, 3x30 min
IP rating	IP67

Interface

Switching output	PNP normally open (NO)
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Material

Housing material	Brass, nickel plated
Material jacket	PVC
Material sensing surface	PMMA
Surface protection	nickel-plated

Mechanical data

Dimension	Ø 18 x 67.5 mm
Mounting part	Nut M18x1
Tightening torque max.	35 Nm

Optical features

Ambient light max.	2000 Lux
Beam characteristic	Divergent
Light type	Infrared
Principle of optical operation	Diffuse sensor, energetic
Switching function, optical	Light-on
Wave length	880 nm

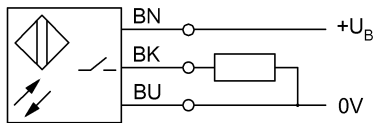
Range/Distance

Hysteresis H max. (% of Sr)	25.0 %
Range	0...400 mm
Rated operating distance Sn	400 mm Adjustable
Repeat accuracy max. (% of Sr)	5.0 %
Temperature drift max. (% of Sr)	10 %

Remarks

Reference object (target): gray card, 200 x 200, 90 % remission, axial approach.
The sensor is functional again after the overload has been eliminated.
Order accessories separately.

Wiring Diagrams



Opto Symbols

